IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2017

SMART GRID TECHNOLOGIES

Friday, 5 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions. All questions carry equal marks

Use answer book A to answer Question 1, answer book B to answer questions 2 and answer book C to answer question 3 & 4.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : S.H. Tindemans, G. Strbac,

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The Questions

1. Power system in Borduria presented in Figure 1.1, with a 2 GW gas-turbine generator operating at the marginal cost of £30/MWh located in the North and 300 MW of peaking generator operating at the marginal cost of £60/MWh located in the South West. Demand in the North is 200MW, in the South West 750MW and in South East 500MW.

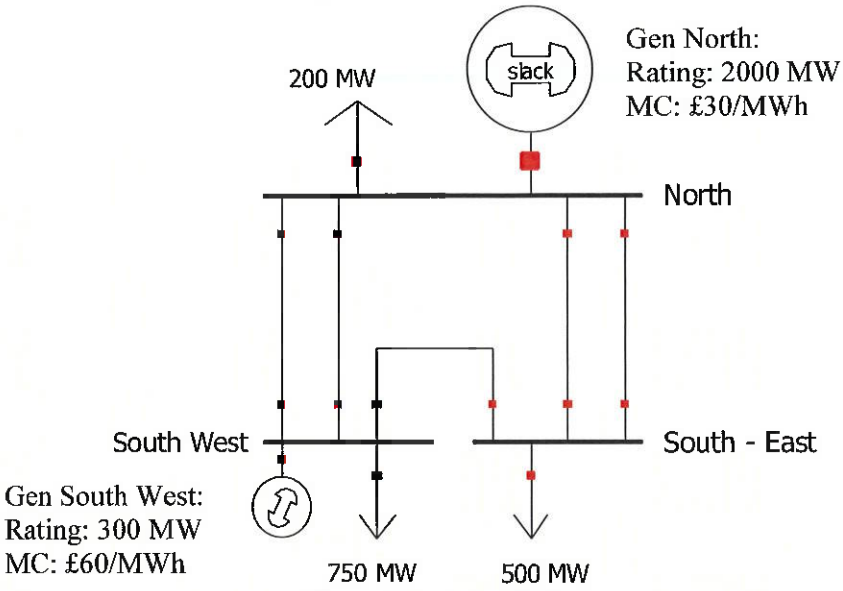


Figure 1.1: A 4-node system

The parameters of the transmission corridors connecting the North and South regions and between the South-East and the South-West region are shown in Table 1.1 below.

Transmission corridor	Number of circuits	X (p.u.)	Capacity per circuit (MW)
North - South East	2	0.50	500
North – South West	2	1.00	375
South East – South West	1	0.25	500

Table 1.1: Parameters of transmission corridors

a) Determine the Economic Dispatch and compute the corresponding power flows in each of the circuits

[3]

b) Determine the power flows following the outage of single individual circuits and identify conditions in which the power flows during contingency exceed the circuit capacity limit.

[3]

c) In case that Demand Side Response may be available to provide corrective control, determine the minimum amount of Demand Side Response that would be needed to manage the power flows within the limits.

[7]

d) If the system operator needs to secure the system against one line outage preventively using the South generator, determine the secured dispatch and the corresponding cost. What is the value of Demand Side Response per each MW made available.

[7]

2. Flexible appliances and big data

a) Describe operational properties and provide illustrations of flexible appliances with continuously adjustable power levels and appliances with shiftable cycles. Provide example of each type explaining their flexibility characteristics.

[3]

b) A simple power system, operating over a two-hour time horizon, includes:

- a generator producing power s_t (MW) at hour t with a cost function $C_t(s_t) = 50s_t^2$ (£) and a maximum output limit $s^{max} = 10\text{MW}$
- inflexible demand appliances, consuming power $D_1 = 4\text{MW}$ (at hour 1) and $D_2 = 3\text{MW}$ (at hour 2)
- 1000 identical flexible demand appliances with power d_t (kW), scheduling interval including hours 1 and 2, total energy required $E = 5\text{kWh}$ and maximum power limit $d^{max} = 6\text{kW}$

i) Assuming that price-based scheduling of the flexible demand appliances is employed, what should be the relation between the prices at hour 1 and 2 (denoted by λ_1 and λ_2 respectively)? What is the resulting power demand of each of the flexible demand appliances? What is the total generation cost?

[4]

ii) Explain how a flexibility limitation approach can be applied to avoid the demand concentration effect in the context of this example. Calculate the most suitable value of the flexibility limit ω minimising the generation cost explaining the rational of your calculation.

[5]

c) Give a brief definition for concept of big data and state the primary sources of big data in power systems.

[2]

d) Give an example of how (i) a transmission system operator (ii) an asset owner (iii) an electricity supplier can use big-data analytics to improve their business performance. For each case, identify a specific task, describe how it was carried out in the past and how it can be improved using the concept of big data.

[6]

3. Reliability models and Monte Carlo simulation

a) A generating unit has failed 20 times in the past 8 years, and repairs have taken 1 week on average. Assuming that this is representative of its overall reliability, what is the mean time to failure (MTTF)? What is the availability of the unit?

[3]

b) Draw the 'bathtub curve' associated with component failures, annotating the different sections of the curve. Discuss how it relates to the continuous time Markov model with exponentially distributed times-to-failure. Note: a mathematical derivation is not required.

[3]

c) Consider a grid-scale storage facility that has three operational states: 'failed', 'fully operational' and 'partially operational'. A Markov model is used to model transitions between states. The transition rates are as follows: (1) from any operational state, complete failures occur at a rate of 1 per year; (2) repairs from the failed state are carried out at a rate of 4 per month and always result in a fully operational unit; (3) fully operational units enter a partially operational state at a rate of 1 per month; (4) partially operational units return to full operation at a rate of 2 per month.

i) Draw the corresponding continuous-time Markov model.

[2]

ii) Draw a discrete time Markov chain for this process, assuming a time step of 1 day. Include all non-zero transition probabilities.

[2]

d) A system operator is designing a forced islanding scheme: in case of an acute blackout threat, the scheme preventively splits the power system into separate electrical islands. Such a scheme must have a high degree of dependability, but also be resilient against spurious activations.

The proposed design is as follows. The state of the power system is monitored at three locations (A, B and C), each equipped with a monitoring station (m_a, m_b and m_c). If a monitoring station detects a critical system condition it sends an alert to the control room over a dedicated communication channel (one for each station – c_a, c_b and c_c). The decision to island the system is taken by a computer in the control room when it receives alert signals from *at least two* sites. To provide redundancy, there are two identical computer systems that receive identical signals and that can each activate the islanding signal.

i) Draw a reliability block diagram for the successful activation of the islanding signal in response to a critical system condition. You may assume that the critical condition is observable at each of the three monitoring stations (*if* they are operational).

[2]

ii) Assuming independence between components and an availability of 95% for monitoring stations, 99% for communication channels and 98% for the computer, quantify the probability that a critical operating condition is successfully detected.

[3]

e) The reliability of a large distribution network is analysed using time-sequential Monte Carlo simulations. In a 2-hour simulation run, 1,000 independent 'simulated years' were generated. The CML index was computed for each year: the average value was 82.1 minutes, and the sample standard deviation was 63 minutes.

i) What is the (estimated) standard error of the average CML value? What would you expect the standard error to be if we were to run the simulation for 24 hours instead of 2 hours?

[3]

ii) State the estimate for the expected CML including its estimated error in three different ways.

[2]

4. Quantification of distribution network reliability

a) Ofgem specifies CML and CI targets for Distribution Network Operators. Discuss how these targets vary across DNOs and regulatory years. Describe the mechanism by which CML and CI targets are used to incentivise network reliability.

[3]

b) Consider a section of a distribution network that serves 8000 customers. Table 4.1 lists the outage events that occurred on this network in the regulatory year 2015/16.

Customers affected	Duration
50	7 hours
3800	90 minutes
1000	1 minute
30	1 day
700	15 minutes
200	8 minutes
1400	4 minutes

Table 4.1: outage events

Compute the CML and CI values for this section of the network.

[3]

c) Consider the distribution network in Figure 4.1.

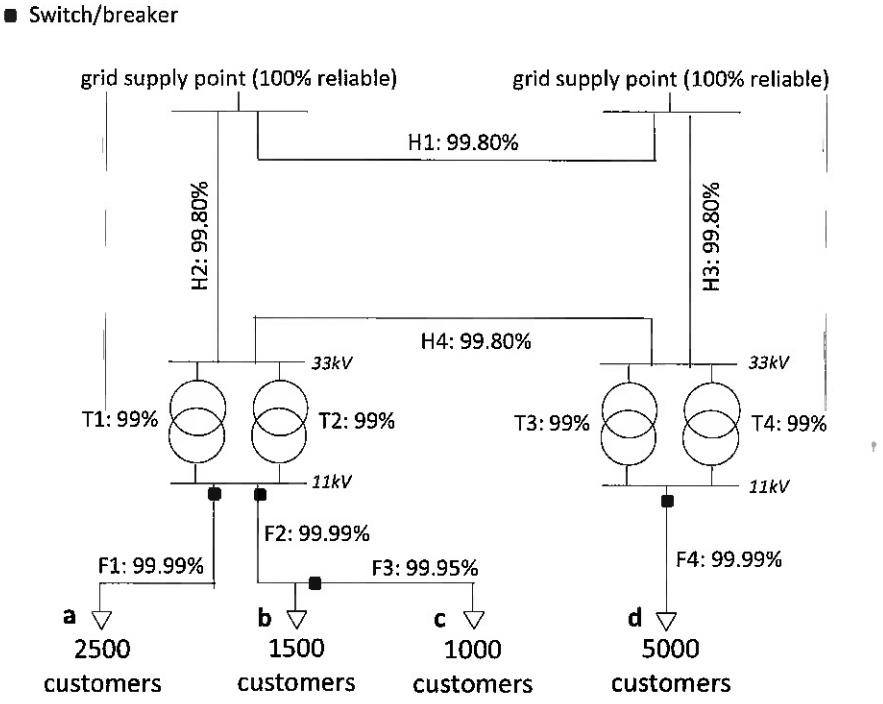


Figure 4.1: network diagram

The network consists of a meshed 33kV network supplied by two grid supply points. There are no constraints on the 33kV network; single lines have sufficient capacity to carry all loads. Customer loads are connected to radial 11kV sections that are supplied through 33-11kV transformers T1-T4. The load points (a-d) are labelled with the number of customers connected to each load point. Component failures are assumed to be independent and to last for more than 3 minutes each. Component availabilities are indicated by percentages alongside each component.

Default load level scenario. The 33-11kV transformers are redundant at all times, i.e. either T1 or T2 can fully supply load points **a-c**, and T3 or T4 can fully supply load point **d**.

i) Compute the probabilities that each of the load points is connected to a grid supply point (one expression for each load point). Then compute the expected CML value for this network.

[6]

Increased load level scenario. Load levels in **d** increase to the point where a single transformer (T3 or T4) is sufficient only 95% of the time. In the remaining 5% of cases, *both* T3 and T4 are required to supply all load. When the available transformer capacity is exceeded, all load in **d** is disconnected.

ii) For the increased load level scenario, compute the probability that loads **d** are supplied. Compute the updated expected-CML value.

[3]

iii) Propose two reasonable engineering solutions to decrease the expected-CML in the increased load level scenario.

[2]

d) State examples of *physical* threats and *cyber* threats to reliable power system operation. For each category, provide one example of an *accidental/natural* threat and one example of a *malicious* threat (i.e. provide four examples in total).

[3]